Attention Hijacking in Vision Transformers: Reproducing Attention-Fool Patch Attacks and Evaluating RSA-Style Defense

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Code: `https://github.com/SujashB/Attention-Fool`

Abstract

This report studies adversarial patch attacks on Vision Transformers, focusing on how dot-product attention can be manipulated by localized patches. Building on Attention-Fool, we reproduce part of the GMYA Table 3 setting by comparing standard cross-entropy PGD patch attacks with attention-based objectives that directly target query-key similarities. We evaluate attacks on ResNet50, ViT, and DeiT classifiers using a 1000-image ImageNet validation subset. Our results partially reproduce the original findings: DeiT models closely match the reported robust accuracies, while ViT models show a larger robustness gap, suggesting sensitivity to checkpoint selection and implementation details. We also evaluate a limited RSA-inspired attention trimming defense on ViT-T. RSA greatly improves robust accuracy under Attention-Fool patch attacks while clean accuracy stays close to undefended ViT-T on the same data and loader, suggesting that the defense mitigates patch attacks without a major clean-accuracy tradeoff in our experiments.

1 Introduction and Motivation

Vision Transformers (ViTs) apply self-attention to image patches. Rather than relying primarily on local convolutional filters, a ViT divides an image into fixed-size patches, embeds those patches as tokens, and uses self-attention to model relationships across the full image [3]. This global reasoning is a major strength of transformer-based vision models: each query token can assign different weights to other key tokens based on query-key similarity [2].

However, the same attention mechanism can also create a vulnerability. Neural networks are known to be sensitive to adversarial inputs, including localized adversarial patches that visibly modify only a small region of the image [4, 5]. In a ViT, such a patch can correspond to one or more image tokens. If the patch token receives high query-key similarity from many other tokens, the model's attention may shift away from useful image content and toward the adversarial patch.

This project reproduces the image-classification attack setting from GMYA [1]. GMYA argues that adversarial patches can exploit dot-product attention directly and proposes Attention-Fool objectives that target pre-softmax attention logits rather than only the final classification loss. We compare standard cross-entropy PGD patch attacks with Attention-Fool variants, including L_{kq}, L^*_{kq}, and momentum-based versions.

Our experiments focus on pretrained ResNet50, ViT, and DeiT classifiers evaluated on a 1000-image ImageNet validation subset. We report clean accuracy and robust accuracy, where lower robust accuracy indicates a stronger attack. We also evaluate a limited RSA-inspired attention trimming defense on ViT-T to test whether suppressing anomalous token influence can reduce the damage caused by strong Attention-Fool patches.

2 Prior Work

Adversarial robustness is central to trustworthy machine learning because models that perform well on clean inputs can fail under carefully designed perturbations. Madry and collaborators formalized adversarial robustness as a robust optimization problem and popularized PGD as a strong first-order attack method [5]. Brown and collaborators introduced adversarial patches, showing that localized visible patterns can cause image classifiers to misclassify inputs even when most of the image remains unchanged [4].

Vision Transformers create a natural setting for patch-based attacks because images are already represented as token sequences. Dosovitskiy and collaborators introduced ViTs by splitting images into fixed-size patches and processing them with transformer layers [3]. Since transformer attention is based on scaled query-key dot products [2], a localized patch can potentially influence the model by changing attention relationships rather than only by changing local image features.

Several works study this vulnerability in transformer-based vision models. Lovisotto and collaborators show that adversarial patches can manipulate dot-product attention and propose Attention-Fool losses that explicitly target query-key similarities [1]. Related work such as Patch-Fool and studies of patch perturbation robustness further show that ViTs are not automatically robust to localized attacks [6, 7]. DeiT models are also relevant because they provide practical pretrained transformer classifiers for this evaluation setting [8].

Prior defenses motivate the defense component of our project. Liu and collaborators study anomalous-score-based defenses for patched attacks on ViTs, while Mu and Wagner propose Robust Self-Attention as a defense against adversarial patches [9, 10]. Our defense is RSA-inspired rather than a full reproduction of Robust Self-Attention: it trims likely anomalous token influence during attention aggregation and is evaluated as a limited exploratory extension.

3 Threat Model

This project studies a white-box inference-time evasion attack against pretrained image classifiers. The adversary's goal is to cause the classifier to mislabel an input image by adding a small adversarial patch at inference time. This is not a training-time poisoning attack or a privacy attack because the model weights are fixed and the adversary modifies only the input image. The primary security asset is prediction integrity: a successful attack causes an image that would otherwise be correctly classified to receive an incorrect label. For Vision Transformers, a secondary concern is the integrity of the internal attention mechanism. If a patch redirects attention away from meaningful image content and toward a malicious patch token, then the attack affects both the final prediction and the model's internal representation process.

We assume that the attacker knows the model architecture, parameters, gradients, and intermediate attention activations. The attacker uses white-box access to optimize a patch with projected gradient descent. The patch does not need to be imperceptible; instead, the attack is spatially constrained. The attacker directly changes only the patch pixels, but those pixel changes can indirectly alter query-key similarities inside dot-product attention layers.

Formally, given an input image x, true label y, adversarial patch p, and fixed patch location L, the patched image is defined as

$$x' = F(x, p, L), \tag{1}$$

where F overwrites the selected image region with the patch. The baseline attack optimizes the patch by maximizing the cross-entropy loss:

$$p^* = \arg\max_p \; \mathcal{L}_{\mathrm{CE}}\big(f(F(x, p, L)), y\big), \tag{2}$$

where f is the pretrained classifier. The Attention-Fool variants extend this objective by adding an attention-targeting term:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{CE}} + \lambda \mathcal{L}_{\mathrm{attn}}, \tag{3}$$

where $\mathcal{L}_{\mathrm{attn}}$ is either L_{kq} or L^*_{kq}. The L_{kq} objective encourages many query tokens to attend to the adversarial patch key token, while L^*_{kq} specifically targets the class token's query in ViT-style classifiers. This distinction is important because the Attention-Fool attack does not only try to increase output loss; it attempts to exploit dot-product attention directly.

In our reproduction, the adversarial patch is 16×16 pixels and is placed in the top-left corner, corresponding to one ViT/DeiT token slot. We evaluate on 1000 ImageNet validation images using pretrained ResNet50, ViT, and DeiT models from `timm` [11, 12]. The attacks are optimized using PGD for 250 iterations with step size $\alpha = 8/255$, cosine decay, and optional normalized momentum with $\beta = 0.9$. We report robust accuracy, where lower robust accuracy indicates a stronger attack. This setup is designed to reproduce part of Table 3 from the GMYA paper by comparing standard cross-entropy PGD attacks against L_{kq}, L^*_{kq}, and momentum variants [1].

This threat model is intentionally controlled. We do not optimize over patch location, model training data, or physical-world transformations such as lighting, camera angle, printing artifacts, or distance. We also do not assume a black-box setting where the attacker only has API access. These restrictions make the experiment suitable for analyzing the core mechanism proposed by GMYA: whether an adversarial patch can exploit dot-product attention by causing model tokens to attend to a malicious patch token. However, the controlled setting also limits real-world generalization.

Finally, our updated implementation highlights an important reproducibility consideration. The threat model matches the original paper, but the realized attack strength can depend heavily on implementation details such as checkpoint selection, `timm` version, and gradient flow through attention modules. Our DeiT results reproduce the paper closely, while the ViT results show a larger gap, likely due to checkpoint differences and/or a ViT-specific attack-path issue. These factors do not change the threat model itself, but they are important for interpreting the experimental results and are discussed later in the report.

4 Methodology

This project reproduces the image-classification experiments of Lovisotto and collaborators by implementing adversarial patch attacks against pretrained CNN and transformer-based image classifiers [1]. The main problem we study is whether a small localized patch can reduce model accuracy either by directly increasing classification loss or by manipulating internal dot-product attention. Our goal is not to train a new model, but to evaluate the robustness of already-trained models under the Table 3-style attack setting from GMYA.

4.1 Reproduction Target

We focus on reproducing representative results from GMYA Table 3. This table compares standard cross-entropy patch attacks against Attention-Fool objectives on ImageNet classifiers. Our reproduction evaluates clean accuracy and robust accuracy for ResNet50, ViT, and DeiT models. Robust accuracy measures the percentage of images still classified correctly after the adversarial patch is applied, so lower robust accuracy indicates a stronger attack.

This reproduction target is important because it lets us test both the attack mechanism and the reproducibility of the original robustness claims. In particular, we compare ordinary PGD patch optimization with Attention-Fool losses that explicitly target query-key similarities inside transformer attention layers.

4.2 Dataset, Models, and Tools

We evaluate on a 1000-image subset of the ImageNet validation set [11]. The patch is 16×16 pixels and is placed in the top-left corner of the image. For ViT and DeiT models with 16×16 patch embeddings, this patch corresponds to one image-token slot, which makes it possible to study whether one malicious token can influence the model globally through self-attention.

The evaluated models are ResNet50, ViT-T, ViT-B, DeiT-T, and DeiT-B. ResNet50 serves as a convolutional baseline because it does not use dot-product self-attention. Therefore, ResNet50 is evaluated only with cross-entropy patch attacks. ViT and DeiT models are the main focus because their image-token representation and self-attention layers match the vulnerability studied in GMYA. Our implementation uses Python, PyTorch, and pretrained models from `timm` version 1.0.27 [12]. The updated reproduction uses PGD for 250 iterations, step size $\alpha = 8/255$, cosine decay, and optional normalized momentum with $\beta = 0.9$.

4.3 Adversarial Patch Formulation

Let x denote an input image, y its ground-truth label, p the adversarial patch, and L the fixed patch location. The patched image is written as

$$x' = F(x, p, L), \tag{4}$$

where F overwrites the selected region of x with the patch p. During the attack, the victim model's parameters remain frozen; only the patch pixels are optimized. The general patch-optimization objective is

$$p^* = \arg\max_p \; \mathcal{L}\big(f(F(x, p, L)), y\big), \tag{5}$$

where f is the victim classifier and $\mathcal{L}$ is the attack loss. After each update, the patch is clipped to the valid image range. Unlike imperceptible perturbation attacks, the patch is allowed to be visible; the constraint is its fixed size and location.

4.4 Baseline Cross-Entropy PGD Attack

The baseline attack uses projected gradient descent (PGD) to maximize the cross-entropy loss at the output of the classifier:

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{CE}}\big(f(F(x, p, L)), y\big). \tag{6}$$

At iteration t, the patch is updated according to

$$p_{t+1} = \Pi_{[0,1]}\left(p_t + \alpha_t \cdot \operatorname{sign}\left(\nabla_p \mathcal{L}_{\text{CE}}\right)\right), \tag{7}$$

where $\Pi_{[0,1]}$ clips the patch to the valid pixel range and α_t is the step size at iteration t. This baseline is necessary because it measures how vulnerable each model is to an ordinary output-level adversarial patch attack before adding attention-specific objectives.

4.5 Attention-Fool Losses

The main contribution of GMYA is the observation that adversarial patches can exploit dot-product attention directly. For layer l and attention head h, let

$$P_Q^{hl} = X^{hl} W_Q^{hl}, \qquad P_K^{hl} = X^{hl} W_K^{hl} \tag{8}$$

be the projected queries and keys. Following GMYA, we normalize projected queries and keys per head and layer so that losses are comparable across heads and layers:

$$\begin{aligned} \bar{P}_Q^{hl} &= \frac{P_Q^{hl}}{\frac{1}{n}\left\|P_Q^{hl}\right\|_{1,2}}, \\ \bar{P}_K^{hl} &= \frac{P_K^{hl}}{\frac{1}{n}\left\|P_K^{hl}\right\|_{1,2}}. \end{aligned} \tag{9}$$

where

$$\|X\|_{1,2} = \sum_{i=1}^{n} \sqrt{\sum_{j=1}^{d_k} X_{ij}^2}. \tag{10}$$

The pre-softmax attention logits used by Attention-Fool are then

$$B^{hl} = \frac{\hat{P}_Q^{hl} (\hat{P}_K^{hl})^\top}{\sqrt{d_k}}, \tag{11}$$

where B_{ji}^{hl} is the dot-product similarity between query j and key i. Let $i^\star$ denote the key token corresponding to the adversarial patch.

For a single head and layer, GMYA defines

$$\mathcal{L}_{kq}^{hl} = \frac{1}{n} \sum_{j=1}^{n} B_{ji^\star}^{hl}. \tag{12}$$

The full $\mathcal{L}_{kq}$ loss combines heads and layers using a smooth maximum, with $\mathrm{smax}(x) = \log \sum_i e^{x_i}$:

$$\mathcal{L}_{kq}^{l} = \log \sum_h e^{\mathcal{L}_{kq}^{hl}}, \qquad \mathcal{L}_{kq} = \log \sum_l e^{\mathcal{L}_{kq}^{l}}. \tag{13}$$

Maximizing this objective encourages many query tokens to attend to the adversarial patch key across the most vulnerable heads and layers.

GMYA also defines a targeted-query variant for architectures with a special class token. Let $j^\star$ be the query index to target, which is the class-token query in our ViT and DeiT experiments. For a single head and layer,

$$\mathcal{L}_{kq^\star}^{hl} = B_{j^\star i^\star}^{hl}, \tag{14}$$

and this loss is generalized over heads and layers in the same way:

$$\mathcal{L}_{kq^\star}^{l} = \log \sum_h e^{\mathcal{L}_{kq^\star}^{hl}}, \qquad \mathcal{L}_{kq^\star} = \log \sum_l e^{\mathcal{L}_{kq^\star}^{l}}. \tag{15}$$

This distinction matters because ViT and DeiT classifiers use the class-token representation for final prediction. If the class token attends strongly to the adversarial patch, the patch may gain direct influence over the model output.

The combined attack objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{attn}}, \tag{16}$$

where $\mathcal{L}_{\text{attn}}$ is either $\mathcal{L}_{kq}$ or $\mathcal{L}_{kq^\star}$, which is reported as L_{kq}^* in our tables. Our implementation evaluates three non-momentum attack objectives: PGD+$\mathcal{L}_{\text{CE}}$, PGD+$\mathcal{L}_{\text{CE}}$+L_{kq}, and PGD+$\mathcal{L}_{\text{CE}}$+L_{kq}^*. In implementation terms, these objectives pull either all query tokens or the class-token query toward the patch key.

4.6 Momentum Variants

We also evaluate momentum-based PGD because GMYA reports stronger attacks when momentum is used. Following the paper, normalized momentum is

$$m^{(t)} = \beta m^{(t-1)} + (1-\beta) \frac{\nabla_p^t}{\|\nabla_p^t\|_2}, \tag{17}$$

and $m^{(t)}$ is used instead of the gradient ∇_p^t in PGD. In our patch update this gives

$$p_{t+1} = \Pi_{[0,1]} \left(p_t + \alpha_t \cdot \mathrm{sign}\left(m^{(t)}\right)\right), \tag{18}$$

where $\beta = 0.9$ and ∇_p^t is the current patch gradient. This gives three momentum-based variants: cross-entropy only, cross-entropy with L_{kq}, and cross-entropy with L_{kq}^*. Including momentum helps determine whether attack strength is limited by the attention objective itself or by unstable patch optimization.

4.7 RSA-Inspired Attention Trimming Defense

In addition to the attack reproduction, we evaluate a limited defense inspired by Robust Self-Attention [10]. The defense assumes that a successful adversarial patch behaves like an anomalous token inside the transformer sequence. This differs from Attention-Fool: the attack changes input pixels to increase patch influence, while the defense changes attention aggregation to reduce the effect of likely anomalous tokens.

We implement ViT-T-RSA by using the same pre-trained ViT-T weights as the undefended model and enabling RSA trim inside every self-attention layer at inference time. The classifier head and all learned weights remain unchanged; only the attention forward pass differs between ViT-T and ViT-T-RSA. After each attention head computes standard scaled dot-product attention and applies softmax to obtain attention weights w, RSA trim is applied to (w, v) before the weighted sum wV. In our setup, the input resolution is 224×224, the ViT patch size is 16×16, and the image-token grid is 14×14. Although the adversarial patch corresponds to a 1×1 ViT token region, we use a 2×2 maximum trim window to include a small margin around the suspected patch token. The model uses 12 transformer layers, 3 attention heads, and the `vit_tiny_patch16_224.augreg_in21k_ft_in1k` checkpoint from `timm`.

RSA trim is applied after the softmax attention weights are computed. Let $w \in \mathbb{R}^{B \times H \times N \times N}$ denote attention weights and $v \in \mathbb{R}^{B \times H \times N \times D}$ denote value embeddings. For ViT-T, $N = 197$, consisting of one class token and 14×14 image tokens. The defense reshapes image tokens into a spatial grid, computes each token's deviation from the mean value embedding, and identifies the region with maximum pooled distance as suspicious. For detected outlier tokens, their value vectors are replaced with the mean value vector and attention weights toward those tokens are replaced with a uniform value. This suppresses abnormal token influence without retraining or adding trainable parameters.

We evaluate RSA on ViT-T across all six attack configurations: PGD+L_{CE}, PGD+L_{CE}+L_{kq}, PGD+L_{CE}+L^*_{kq}, and their momentum variants. Because the defense is tested only on ViT-T and is not tuned across models or patch settings, it should be interpreted as an exploratory defense experiment rather than a complete mitigation.

5 Experimental Results

We evaluated the reproduced Attention-Fool attack configurations on pretrained ResNet50, ViT, and DeiT classifiers using a 1000-image ImageNet validation subset. For each model, we measured clean accuracy and robust accuracy under different adversarial patch attacks. Robust accuracy is the percentage of images still classified correctly after the adversarial patch is applied; therefore, lower robust accuracy indicates a stronger attack. The implemented attack configurations include PGD with cross-entropy loss, PGD with cross-entropy plus L_{kq}, PGD with cross-entropy plus L^*_{kq}, and the corresponding momentum variants. The reproduction uses a 16×16 top-left patch, 1000 ImageNet validation images, PGD for 250 iterations, step size $\alpha = 8/255$, optional normalized momentum with $\beta = 0.9$, and pretrained models from `timm`.

5.1 Clean Accuracy

Clean accuracy was measured before applying any adversarial patch. This provides a baseline for checking whether the loaded models behave similarly to the original GMYA setup before attack. The reproduced clean accuracies are shown in Table 1.

Table 1: Clean accuracy before applying adversarial patches.

Model	Clean Accuracy
ResNet50	79.60%
ViT-T	73.50%
ViT-B	83.80%
DeiT-T	71.00%
DeiT-B	81.60%

These clean accuracies show that the models were functioning normally before attack. In particular, the DeiT clean accuracies are close to the values reported in GMYA Table 3, and the ViT-B clean accuracy is also reasonably close to the paper's reported ViT-B clean accuracy. This matters because it suggests that later differences in robust accuracy are not simply caused by broken model loading or poor clean performance.

5.2 Robust Accuracy Across Attention-Fool Configurations

Figure 1 compares robust accuracy across three non-momentum attack configurations: the cross-entropy PGD baseline, the L_{kq} attention variant, and the L^*_{kq} attention variant. The results show that attention-targeting losses improve attack strength in several cases, but the improvement is not uniform across all models.

For DeiT-T, robust accuracy decreases from 20.3% under PGD+$\mathcal{L}_{\mathrm{CE}}$ to 18.5% with L_{kq} and 18.2% with L^*_{kq}. For DeiT-B, robust accuracy decreases from 38.0% under PGD+$\mathcal{L}_{\mathrm{CE}}$ to 35.2% with L_{kq} and 35.1% with L^*_{kq}. These

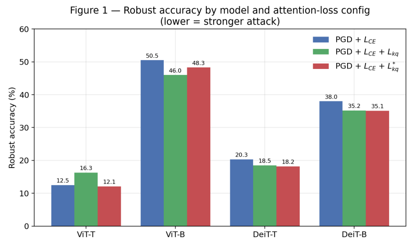

Figure 1: Robust accuracy by model and attention-loss configuration. Lower robust accuracy indicates a stronger attack.

results support the GMYA idea that directly targeting attention can strengthen adversarial patch attacks.

The ViT results are less consistent. For ViT-B, L_{kq} improves the attack by reducing robust accuracy from 50.5% to 46.0%, but L^*_{kq} is weaker at 48.3%. For ViT-T, L^*_{kq} is slightly stronger than the baseline, reducing robust accuracy from 12.5% to 12.1%, while L_{kq} makes the attack weaker at 16.3%. This suggests that Attention-Fool losses are sensitive to the model checkpoint, implementation details, and optimization path.

5.3 Effect of Momentum

Figure 2 isolates the effect of momentum by comparing PGD+$\mathcal{L}_{\text{CE}}$ against PGD+$\mathcal{L}_{\text{CE}}$+Momentum. Momentum substantially strengthens the attack for most transformer models.

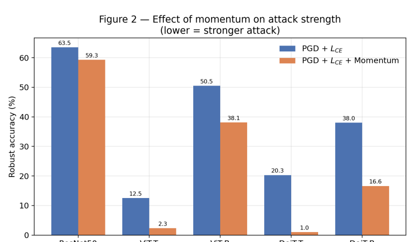

Figure 2: Effect of momentum on attack strength. Lower robust accuracy indicates a stronger attack.

The largest improvement occurs for DeiT-B, where robust accuracy drops from 38.0% to 16.6%, a decrease of 21.4 percentage points. DeiT-T also drops sharply from 20.3% to 1.0%. ViT-T drops from 12.5% to 2.3%, and ViT-B drops from 50.5% to 38.1%. ResNet50 shows a smaller decrease, from 63.5% to 59.3%. These results show that the optimization procedure itself has a large effect on attack strength. Momentum appears especially important for transformer models, likely because it stabilizes the patch update direction across PGD iterations and helps the attack find more damaging perturbations.

5.4 Comparison with GMYA Table 3

Figure 3 compares our PGD+$\mathcal{L}_{\text{CE}}$ robust accuracy values against the corresponding values from GMYA Table 3. This figure highlights the main reproducibility finding: DeiT reproduces closely, while the ViT family diverges.

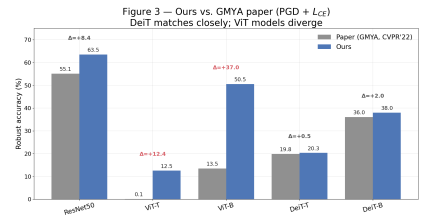

Figure 3: Ours vs. GMYA under PGD+$\mathcal{L}_{\text{CE}}$. DeiT matches closely, while ViT models diverge.

For DeiT-T, the paper reports 19.8% robust accuracy under PGD+$\mathcal{L}_{\text{CE}}$, while our result is 20.3%, a difference of only 0.5 percentage points. For DeiT-B, the paper reports 36.0%, while our result is 38.0%, a difference of 2.0 percentage points. These results suggest that the DeiT portion of the reproduction is successful.

The ViT results differ much more. For ViT-T, the paper reports 0.1%, while our result is 12.5%. For ViT-B, the paper reports 13.5%, while our result is 50.5%. The largest discrepancy is ViT-B, where our attack is 37.0 percentage points weaker than the paper. This discrepancy is important because it appears even in the plain PGD+$\mathcal{L}_{\text{CE}}$ baseline. Since this baseline does not use L_{kq} or L^*_{kq}, the ViT gap cannot be explained only by an incorrect Attention-Fool loss implementation. Instead, it suggests a broader reproducibility issue involving ViT checkpoint selection, software version differences, or attack-path behavior.

5.5 Visual Example of the Patch Constraint

Figure 4 shows an example clean image and the same image with the 16×16 adversarial patch inserted in the top-left corner. The patch is visible but spatially small. This illustrates the key constraint of the adversarial patch threat model: the attacker does not modify the entire image, but instead optimizes a localized region.

For ViT and DeiT models using 16×16 patch embeddings, this localized patch corresponds to one image-token slot. This makes the attack especially relevant to

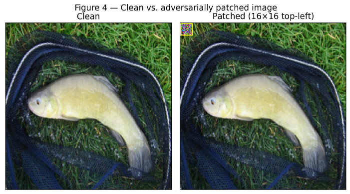

Figure 4: Clean image compared with an adversarially patched image using a 16×16 top-left patch.

transformer-based vision models because a single malicious token can potentially influence the model globally through self-attention.

5.6 Attention Visualization

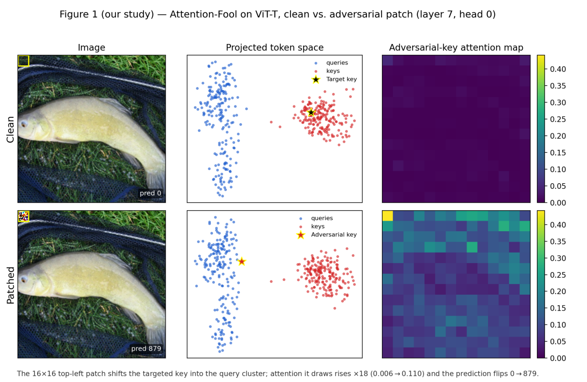

Figure 5: Attention-Fool mechanism on ViT-T. The 16×16 top-left patch shifts the adversarial key toward the query cluster, increases attention toward the patch token, and flips the prediction from class 0 to class 879.

Figure 5 provides qualitative evidence for the Attention-Fool mechanism in our reproduction. The adversarial patch changes both internal attention behavior and final model output, supporting the GMYA claim that adversarial patches can manipulate query-key dot-product attention, not only final classification loss.

5.7 RSA-Inspired Defense Results

We also evaluated an RSA-inspired attention trimming defense on ViT-T across all six attack configurations. Compared with the undefended ViT-T model, ViT-T-RSA achieves much higher robust accuracy under every attack. For example, under PGD+L_{CE}+Momentum, robust accuracy increases from 2.3% to 44.3%. For the two strongest momentum Attention-Fool variants, robust accuracy increases from 4.3% to 37.6% and from 2.6% to 38.0%, respectively.

Table 2: RSA-inspired defense results on ViT-T.

Attack	ViT-T Robust Acc.	ViT-T-RSA Robust Acc.
PGD+L_{CE}	12.5%	46.5%
PGD+L_{CE}+L_{kq}	16.3%	46.2%
PGD+L_{CE}+L^*_{kq}	12.1%	46.6%
PGD+L_{CE}+Mom	2.3%	44.3%
PGD+L_{CE}+Mom+L_{kq}	4.3%	37.6%
PGD+L_{CE}+Mom+L^*_{kq}	2.6%	38.0%

ViT-T-RSA greatly improves robust accuracy under Attention-Fool patch attacks while clean accuracy stays close to undefended ViT-T on the same data and loader, so the defense mitigates patch attacks without a major clean-accuracy tradeoff in our experiments.

6 Discussion of Results

The results support the main GMYA claim: a small 16×16 localized patch can sharply reduce robust accuracy. The effect is strongest on transformer models, with ViT-T dropping from 73.5% clean accuracy to 12.5% under PGD+$\mathcal{L}_{\mathrm{CE}}$ and 2.3% with momentum. DeiT-T and DeiT-B also become highly vulnerable under the strongest momentum Attention-Fool attacks, reaching 0.2% and 13.7% robust accuracy, respectively.

Attention-Fool losses help most clearly on DeiT, but the ViT results are less consistent. Momentum is a major factor across models, reducing robust accuracy from 12.5% to 2.3% for ViT-T, 20.3% to 1.0% for DeiT-T, and 38.0% to 16.6% for DeiT-B. This suggests that patch optimization details can matter as much as the attention loss itself.

The key reproducibility finding is that DeiT matches GMYA closely, while ViT diverges. Under PGD+$\mathcal{L}_{\mathrm{CE}}$, our DeiT-T and DeiT-B results differ from the paper by only 0.5 and 2.0 percentage points, while ViT-B differs by 37.0 percentage points. The RSA results are encouraging: ViT-T-RSA improves robust accuracy across all six attacks while clean accuracy stays close to the undefended baseline on the same data and loader.

7 Limitations

This project has several limitations. First, the reproduction uses a controlled white-box digital attack setting with a fixed 16×16 patch in the top-left corner. This setting is useful for analyzing the GMYA mechanism, but it does not evaluate physical-world conditions such as lighting changes, camera angle, printing artifacts, distance, or patch placement variation. The dataset is also limited to a 1000-image ImageNet validation subset rather than the full validation set, so the results should be interpreted as a representative reproduction rather than a complete benchmark.

Second, the experiments were constrained by limited compute resources and time. Each attack requires many forward and backward passes through large pretrained models, and the full experimental grid across models, attack losses, momentum settings, checkpoints, patch locations, and defense variants is computationally expensive. Because of these constraints, we were not able to fully implement and evaluate the defense component across all models. Instead, we treat the RSA-inspired defense as a limited exploratory direction and focus the main results on reproducing and analyzing the GMYA attack setting.

Third, the reproduction is sensitive to implementation details. Our DeiT results closely match GMYA, but the ViT results show a large robustness gap despite similar clean accuracy. This suggests that checkpoint selection, `timm` version, preprocessing, or gradient flow through attention modules can significantly affect adversarial robustness results. Therefore, our ViT results should not be interpreted as disproving GMYA, but rather as evidence that adversarial robustness reproduction requires exact checkpoint and software control.

The defense evaluation is also limited. RSA-inspired trimming was evaluated only on ViT-T, and although it improves robust accuracy across all six attacks while keeping clean accuracy close to the undefended baseline on the same data and loader, it has not been tuned or stress-tested broadly. Due to time and compute limits, we could not tune RSA hyperparameters, test alternative trimming sizes, evaluate larger models, or run broader adaptive defense-aware attacks. The defense results should therefore be interpreted as preliminary evidence of a promising mitigation rather than a complete defense.

8 Conclusion

This project reproduced and analyzed Attention-Fool-style adversarial patch attacks on CNN and transformer-based image classifiers. The results show that a small 16×16 localized patch can substantially reduce robust accuracy, especially for ViT and DeiT models. Momentum was one of the strongest factors in attack success, and Attention-Fool losses improved attacks on DeiT models but behaved less consistently on ViT models.

The reproduction also revealed an important robustness reproducibility issue. DeiT results closely matched GMYA Table 3, while ViT results diverged significantly despite similar clean accuracy. This suggests that adversarial robustness evaluations depend heavily on exact checkpoints, library versions, preprocessing, and gradient flow through model wrappers.

Finally, the RSA-inspired defense experiment showed that attention trimming can blunt Attention-Fool attacks on ViT-T, improving robust accuracy across all six attack configurations while keeping clean accuracy close to the undefended ViT-T baseline on the same data and loader. Overall, the project supports the GMYA claim that dot-product attention can be exploited by adversarial patches, while also showing that reliable robustness evaluation and practical defense design require careful implementation control.

References

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